Reproducing Evolutionary Suicide in the Life Engine

Oct 16, 2026 · @Ashley

Summary

We achieved evolutionary suicide in the unmodified Life Engine, through an evolved plant-killing trait. We did not achieve it in our modified savanna, where we got failure to adapt and predator boom-bust instead.

- **Part 1, base engine (18 runs).** Killer cells hurt everything in the original engine, so zebras that carried a kick destroyed every plant and most of their own herd, and died out by tick 8,000 in every such condition. When zebras had to evolve the killer from nothing, one seed met all four of our suicide criteria: the killer spread, the herd peaked at 528, every plant was destroyed, and zebras died out at tick 14,005. Two more seeds collapsed on their own but kept their grass. The base engine could not test our grazing hypothesis, because mouths cannot hurt plants and nothing starves.
- **Part 2, modified engine (97 runs, including 18 follow-ups of up to 200,000 ticks).** We added grazing, worse bush food, hunger, meat-eating leopards, our own sprint rule, the four member cells (horn, stripe, cache, soil water), a river, trees and fire. Rhinos failed to adapt in every main seed, and leopards boomed and crashed every time. Zebras evolved a second mouth and survived where they did, but grass never collapsed because of it, so the overgrazing suicide we hypothesised did not happen.
- **Environment mattered as much as traits.** Removing the river or soil water killed every zebra lineage by tick 40,000, against 1 in 5 in the full design.

Question and evidence rule

We asked whether a savanna of zebras, rhinos, leopards, grass and bushes can drive itself extinct by mis-adaptation, and how that differs from failure to adapt.

- **Evolutionary suicide:** a trait spreads because it helps individuals, and the same trait removes something the population depends on, so the population collapses.
- **Failure to adapt:** the species lacks the traits it needs and declines without first succeeding.

As we committed to in Assignment 3, a decline alone does not count as suicide. The analysis script (`analyze.py`) labels a collapse as evolutionary suicide only when all four of these hold:

1. The population grew to at least 1.5 times its starting size.
2. A trait spread before the decline: mouths per animal rose by 0.4 or more, or the share with a killer cell rose by 0.3 or more.
3. The food plants (grass plus bush organisms) fell by at least 25% between the peak and extinction.
4. The decline was self-caused (starvation, old age without replacement, or killing by the lineage's own evolved killers) rather than caused by another species.

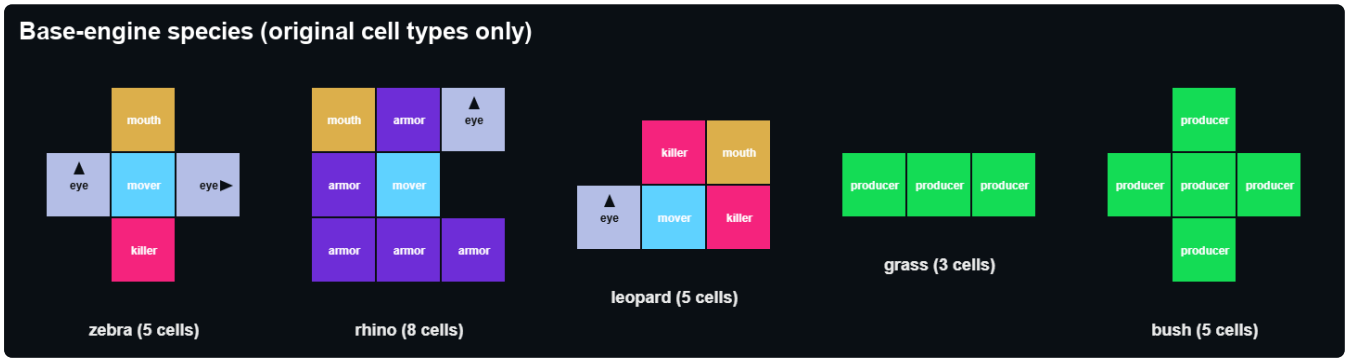
Three of four with growth and trait spread is reported as a suicide candidate. Collapses without growth are failure to adapt. Collapses where other species did most of the killing are collapse under predation. For leopards we separate boom-bust from over-predation (the prey also crashed) and boom-bust from a cohort ageing out.

Species

Each species copies a real animal or plant's strengths and weaknesses. The base engine and the modified engine use different versions, because the base engine has none of our new cells and treats movers differently.

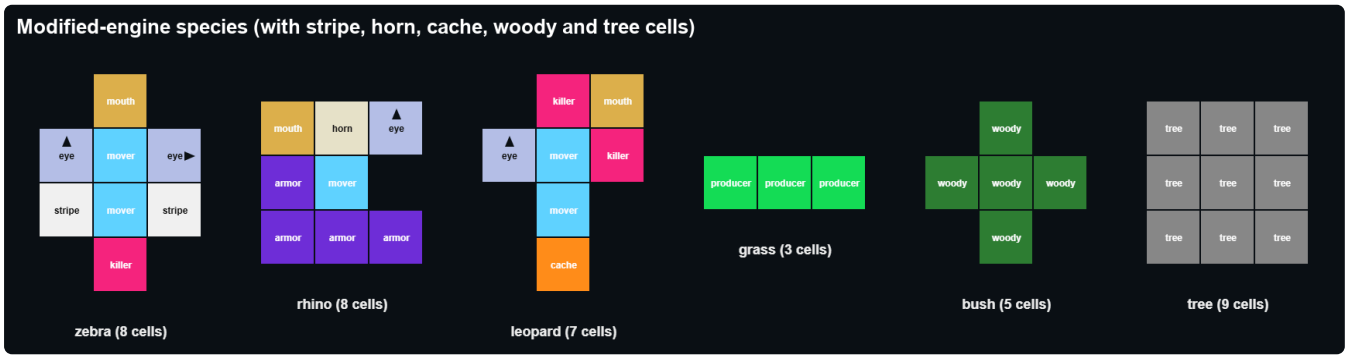
About mover cells. In the original Life Engine an organism needs only one mover cell, and its creator states that adding more does not make it faster or give it anything special: every mover moves one cell per tick. So the base zebra and leopard have one mover each. In the modified engine we added our own rule, explained under Modifications, that lets each extra mover add a step per tick while chasing or fleeing. The modified zebra and leopard carry two movers to use it.

Base-engine species (original cells only)



Species	Cells	Real-world basis	Notes
Zebra	1 mover, 1 mouth, forward eye, side eye, rear kick (killer) (5)	Herd grazer that kicks predators	In the original engine the kick hurts anything it touches, including other zebras and plants
Rhino	1 mover, 1 mouth, forward eye, 5 armour (8)	Huge, armoured grazer	Armour only blocks killer cells; no extra health or cost
Leopard	1 mover, front killer and side killer beside the mouth, eye (5)	Ambush hunter	Moves no faster than its prey; can also eat plant food
Grass	3 producers in a row	Fast-spreading grass	Same food as bush
Bush	5 producers in a plus	Larger plant	Same food as grass

Modified-engine species



Species	Cells	Strengths	Weaknesses
Zebra	2 movers, 1 mouth, forward eye, side eye, rear kick, 2 stripes (8)	Sprint (our rule); herd camouflage; kick wounds leopards	One mouth; bush food worth 0.35; no armour
Rhino	1 mover, 1 mouth, horn, forward eye, 4 armour (8)	Horn immune to killers and gores attackers; armour adds 3 health per cell (20 in total)	Slow; armour costs 2 extra food per calf (16 per calf); poor on bush food

Species	Cells	Strengths	Weaknesses
Leopard	2 movers, jaws (front and side killer beside the mouth), side eye looking forward, cache (7)	Sprint (our rule); bite does 4 damage; eats its kill on the spot; banks surplus near trees	Meat only; no armour; gored by horns, kicked by zebras
Grass	3 producers in a row	Reproduces fastest; full-value food	Thirsty; can be grazed
Bush	5 woody cells in a plus	Cannot be grazed; uses half the water	Browse feeds grazers poorly
Tree	3x3 tree cells	Lives 6 times longer; resists fire; hides animals beside it	Very slow; thirsty

Starting numbers in every run: 140 grass, 45 bush, 14 trees (modified only), 36 zebras in 6 herds, 12 rhinos and 16 leopards, on a 160 x 100 grid. Zebras start without a flight response; fleeing can evolve through brain mutation.

Part 1: the base engine

In the unmodified engine, killer cells destroyed the plants and the zebras' own herd every time zebras carried a kick. When the killer had to evolve from nothing, it produced one clear evolutionary suicide and two near misses. The engine could not test our grazing hypothesis at all.

What the base engine can and cannot do

The base runs use the original rules only, plus our Assignment 2 change that lets plants reproduce from the food they make. Every new setting stays off, and mutation cannot create our new cell types.

The base engine allows	The base engine cannot do
Killer cells hurt anything they touch (plants, other species and their own kind), and a dead organism becomes food. A killing trait is rewarded with a meal, while each plant killed is a producer gone for good	Starvation: animals die only of old age or killer damage, so scarcity only acts through failure to breed

The base engine allows	The base engine cannot do
Uneaten food blocks births, so food can choke a population	A worse food: grass and bush drop identical food
Mutation can add, change or remove cells, including killers	Predator dependence: leopards can eat plant food, and carcasses become plant food
	Overgrazing: mouths cannot hurt plants, and eating frees space so plants produce more
	Speed differences: every mover moves one cell per tick, and extra movers do nothing

How we tried to get suicide or failure to adapt

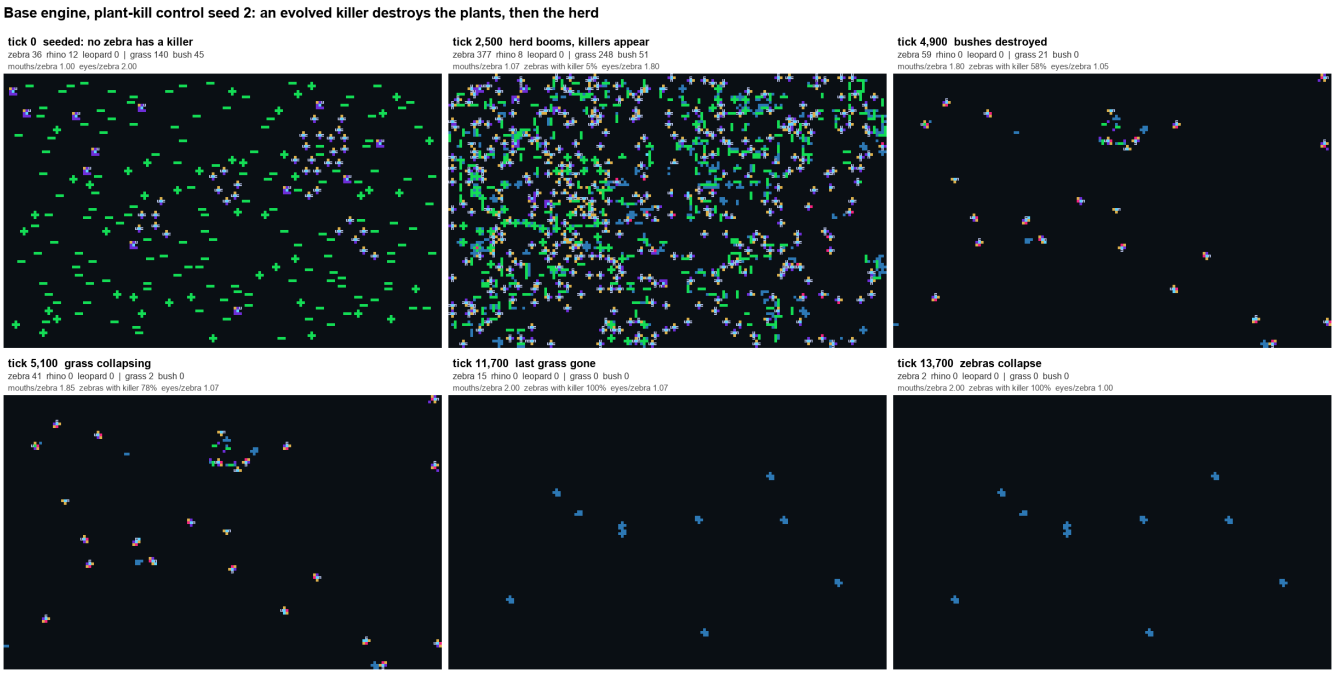
Each condition ran for 3 seeds of 40,000 ticks.

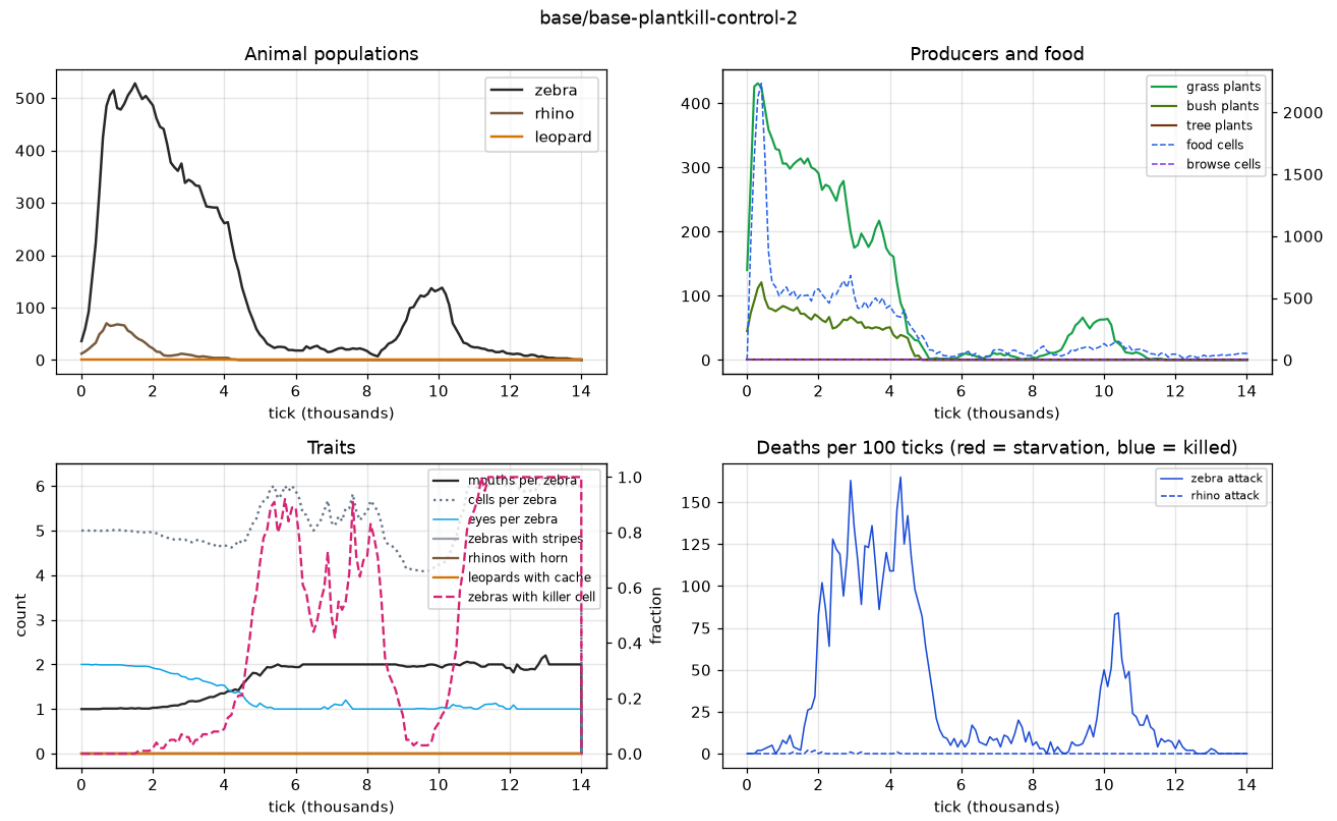
Condition	Aim
Default	The original settings as shipped (food production 5, lifespan multiplier 100, food blocks births). Does anything collapse on its own?
Plant-kill control (route 1)	Zebras start without a killer cell. Insta-kill on, killers reach all 8 neighbours, food production 3. If a killer evolves, each plant it destroys is a meal, so the trait should spread and then remove the producers
Plant-kill, seeded	The same, but zebras start with the kick: the trait is given, not evolved
Harsh	Food production 2: does a species fail, or adapt?
No mutation	Food production 2 and mutation off: the failure-to-adapt baseline, where nothing can change
Clogging (route 2)	Fast plants (food production 10, growing into 8 neighbours) with few grazers: can plants choke on their own food?

What happened, and why

Evolutionary suicide when the killer had to evolve. In the plant-kill control, zebras started without a killer. In seed 2 one evolved and spread to 89-100% of the herd. The herd had peaked at 528, the killers destroyed every bush by tick 4,900 and every grass plant by tick

11,700, and the zebras died out at tick 14,005. This meets all four criteria.



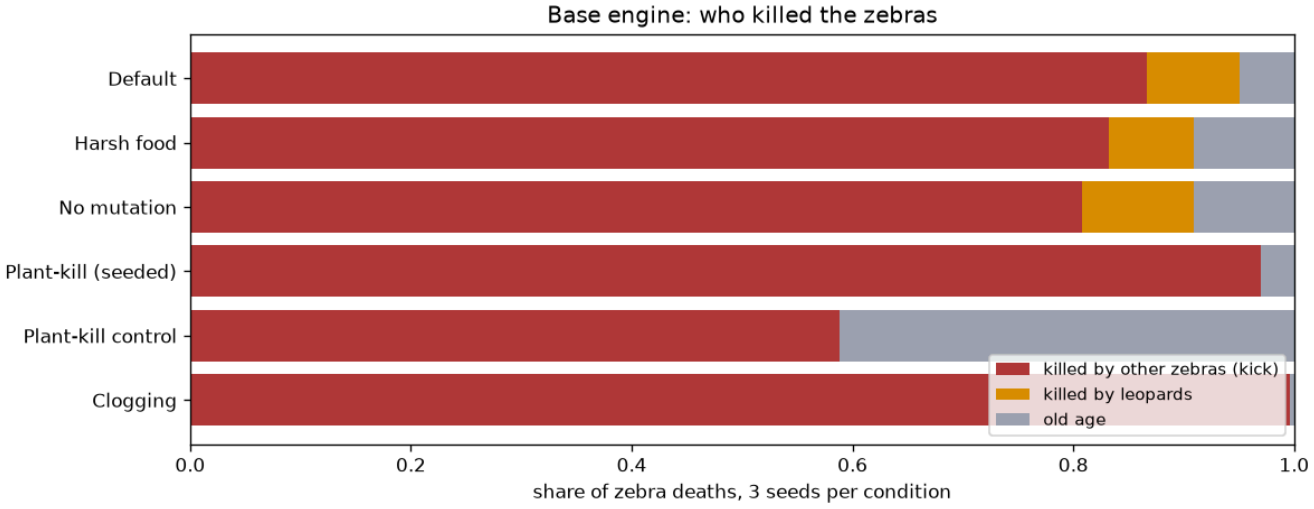


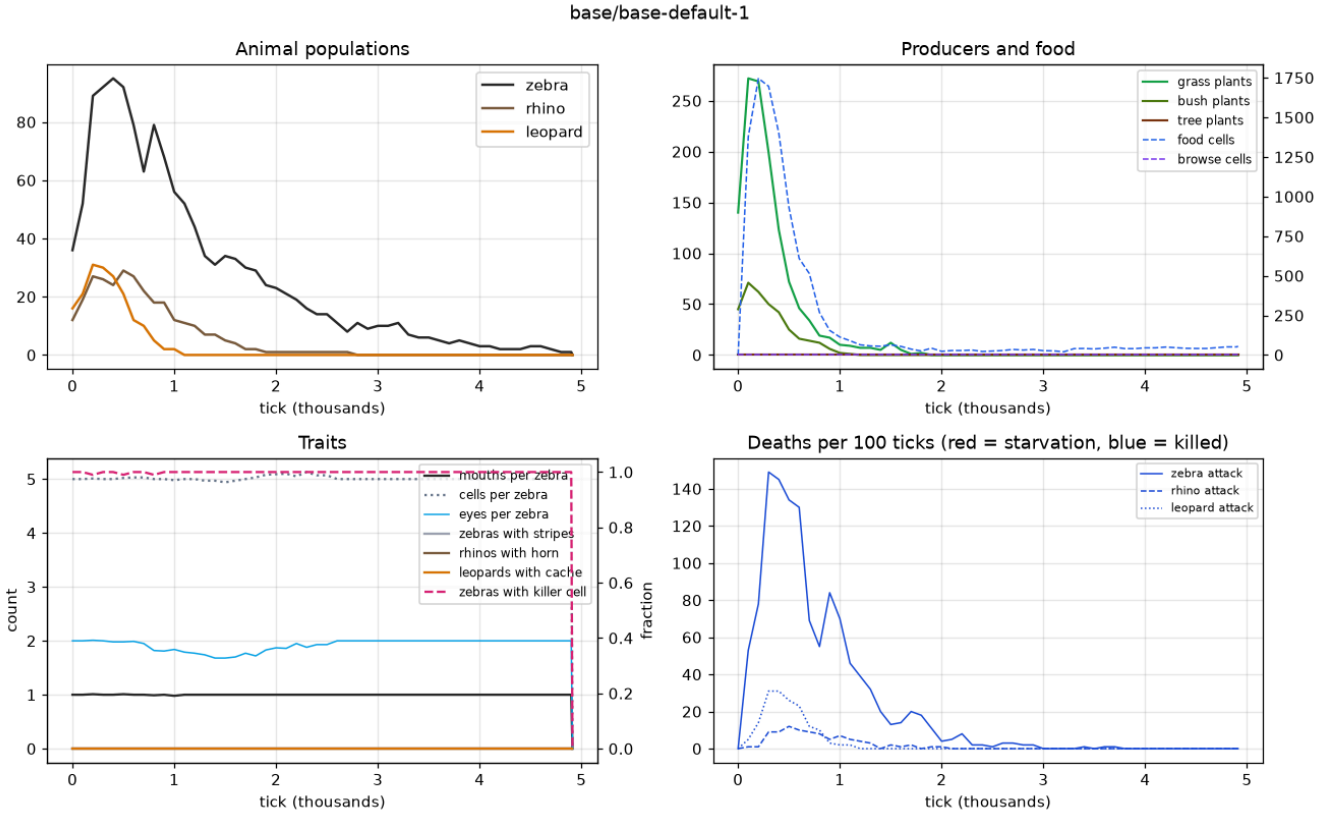
In seeds 1 and 3 (peaks of 594 and 582), the evolved killers mostly killed other zebras rather than plants. Each carcass became food, food piled up to 5,000-7,400 cells, and because uneaten food blocks births, the herd stopped breeding. It died out at ticks 28,000 and 11,850 while grass survived. These are self-made collapses, but they miss the "food plants fell" criterion, so they count as suicide candidates (3 of 4).

Self-destruction by a seeded kick. In every condition where zebras started with the kick (default, harsh, no mutation, seeded plant-kill):

- All bushes were gone by tick 500-1,500 and all grass by tick 700-6,100.
- Zebras died out by tick 2,800-8,000.
- Other zebras' kicks caused 81-97% of zebra deaths; leopards caused 0-10%.
- Kicks also killed most rhinos (69 of 90 rhino deaths in default seed 1). Rhinos were extinct by tick 600-2,800.

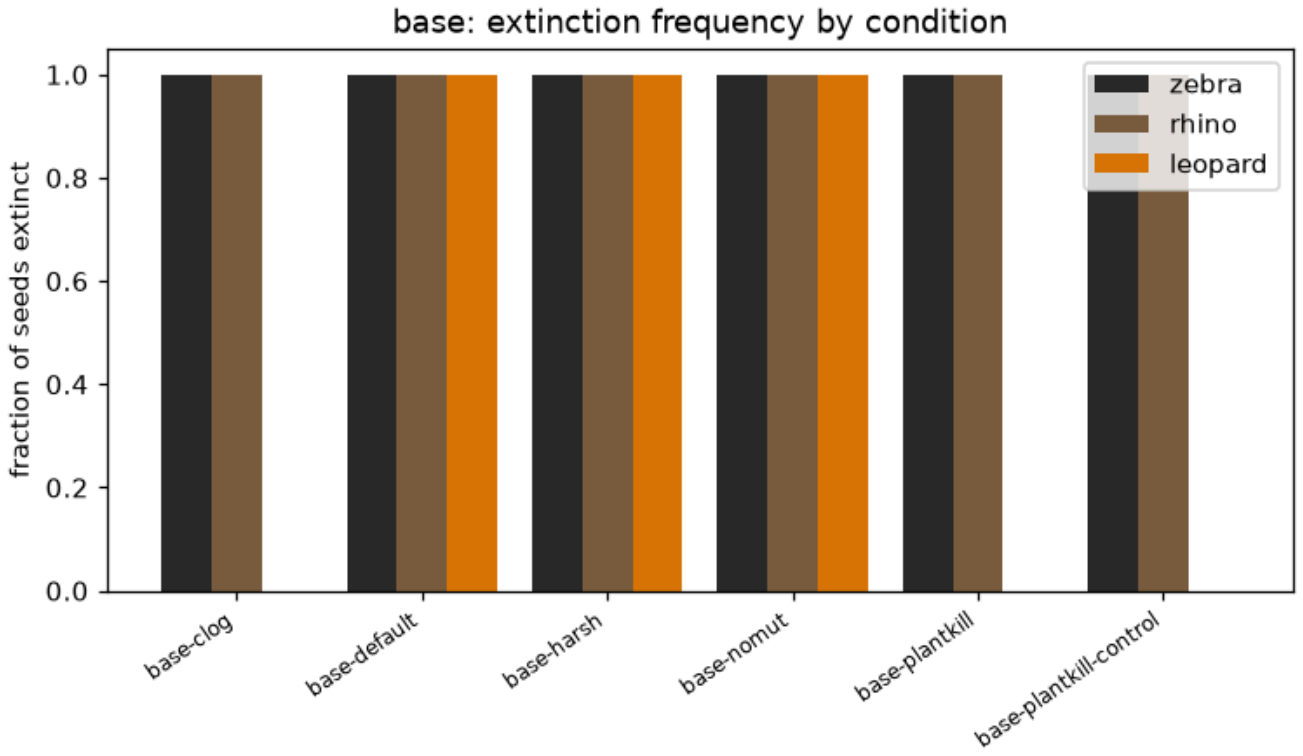
The seeded plant-kill runs had no leopards, so zebra kicks alone were enough. Because the trait was given rather than evolved, these runs fail the trait-spread criterion. They show a design that destroys its own world, not evolutionary suicide.





Failure to adapt.

- With mutation off, zebras died out by about tick 5,000 in all seeds, with no possible change.
- Leopards in every base condition rose to 22-35 and died out by tick 1,300. They moved no faster than their prey and could live on plant food only while plants lasted.
- In the clogging runs, grass filled the map (about 2,000 plants) and plants never choked themselves. The few zebras boomed to 242-348 on the abundant food, then kicked each other to death (other zebras caused all zebra deaths), dying out by tick 8,700-30,600. Route 2 failed: grazers, not plants, were the weak point.



Why the base engine could not test our hypothesis. Zebras cannot overgraze because mouths do not hurt plants. Bush food is the same as grass food. Nothing starves, and leopards do not need prey. The only road to collapse in the base engine is killer cells. That is why we built the modifications in Part 2.

Part 2: what we changed in the engine

Each change answers one of the base-engine limits from Part 1 and adds an escape, so collapse becomes possible but never automatic. All of them default to off, which is why the base runs reproduce the original engine.

Base-engine limit (Part 1)	Change that answers it
Mouths cannot hurt plants, so no overgrazing	Grazing damages grass plants
Grass and bush food are identical	Bush browse worth 0.35 of grass food
Nothing starves	Hunger
Leopards eat plants, carcasses become plant food	Meat, jaws and feeding at the kill
Every animal moves one cell per tick; extra movers do nothing	Our own sprint rule (see below)
Killers hurt their own herd and plants	Killers no longer hurt plants or their own lineage; weapons only strike predators
Plant growth is unlimited	Soil water and fertility, river, trees, fire

Sprinting is our invention, not a Life Engine feature. In the original engine extra mover cells add nothing. We tested leopards at equal speed and they almost never caught zebras, so we added a rule: each extra mover gives one extra step per tick, but only while the animal is chasing or fleeing another organism. Grazing and wandering stay at one step per tick, so fast animals do not trample their own food.

Member features

Feature	Owner	Cell or file	Opens a route to collapse	Escape that makes it hard
Horn cell	Vimbai	`HornCell.js`	Horned rhinos trample nearby plants and food (5% per tick per neighbour)	Horn is immune to killers and gores attackers (50% per tick), so rhinos survive predation

Feature	Owner	Cell or file	Opens a route to collapse	Escape that makes it hard
Stripe cell	Moreblessing	`StripeCell.js`, `EyeCell.js`	More surviving zebras put more pressure on grass	A leopard's eye misses a striped zebra with probability 25% per striped neighbour within 4 cells, up to 85%
Cache cell	Ashley	`CacheCell.js`, `Organism.js`	Banking food delays breeding	Surplus (at least half a litter's food) is banked, up to 3 units per cache cell, near a tree; it is spent automatically when hunger would kill
Soil water and fertility	Fourth member	`ProducerCell.js`, `WoodyCell.js`, `WorldEnvironment.js`	Grass needs water to produce (0.5 per event), so overgrazed, dry soil recovers slowly	Rain refills soil (0.2% per tick towards a Perlin-noise capacity); deep-rooted bushes use half as much water and expand when grass declines

Shared ecology needed to test the hypothesis

Change	Why it was needed
Bush food (browse) worth 0.35 of grass food	Gives zebras a worse fallback food, the core of our hypothesis
Grazing: a mouth next to a grass plant damages it (2% per tick)	Lets mouths deplete plants, which the base engine cannot do
Hunger: pay 1 food every 80 ticks; die after 10 missed payments in a row	Makes scarcity lethal

Change	Why it was needed
Carcasses become meat; jaws (a killer beside a mouth) digest meat only; a leopard eats its kill on the spot	Makes leopards depend on prey
Sprint (our own rule; in the original engine extra movers do nothing): each extra mover adds a step per tick, only when chasing or fleeing	Lets predators catch prey without fast animals trampling their own food
Armour costs 2 food to grow and adds 3 health	Makes rhinos tough but slow to breed, as real rhinos are
Weapons (a killer away from the mouth, like the zebra kick) do 2 damage and only strike predators	Stops zebra kicks from killing rhinos (found and fixed during testing)
River: impassable 3-cell channel; only animals of 9 or more cells can wade; riverbank soil refills 3 times faster	Permanent green refuge; splits the map
Trees: long-lived 3x3 plants that hide nearby animals from the other diet (60%)	Cover for prey and ambush for leopards; the only place caches fill
Fire: random fires (more likely in the dry season) spread through dry vegetation; wet ground and gaps stop them	Grazed land burns less, so diversity changes the environment
Eco logger (`EcoStats.js`) and harness	Records population, traits, food, water and deaths by cause every 100 ticks, and saves screenshots at fixed ticks and at every collapse

How the features connect

1. Rain and soil water set how fast grass regrows. Zebras and rhinos graze it, and trampling by horned rhinos adds to the damage.
2. When grass falls, water is freed for bushes, which expand and offer worse food.
3. Leopards hunt zebras and rhinos. Stripes, tree cover, the kick and the horn reduce kills, and the cache lets leopards wait out lean periods.
4. The river keeps a strip of wet, fertile ground that only rhinos can reach across. Fires remove dry vegetation, and grazed ground acts as a firebreak.

No hard-coded outcomes

No line of code sets a population, a death or a survival. Every death comes from the same per-organism rules: old age, damage from cells, or hunger. The variant worlds only change starting organisms and settings (`savanna-experiments/make-world.js`). Outcomes differ between seeds of the same condition, which a scripted result could not do.

Modified engine: did we achieve it?

Not evolutionary suicide. We did get failure to adapt, and predator boom-bust. The sections below give the evidence; this is the verdict.

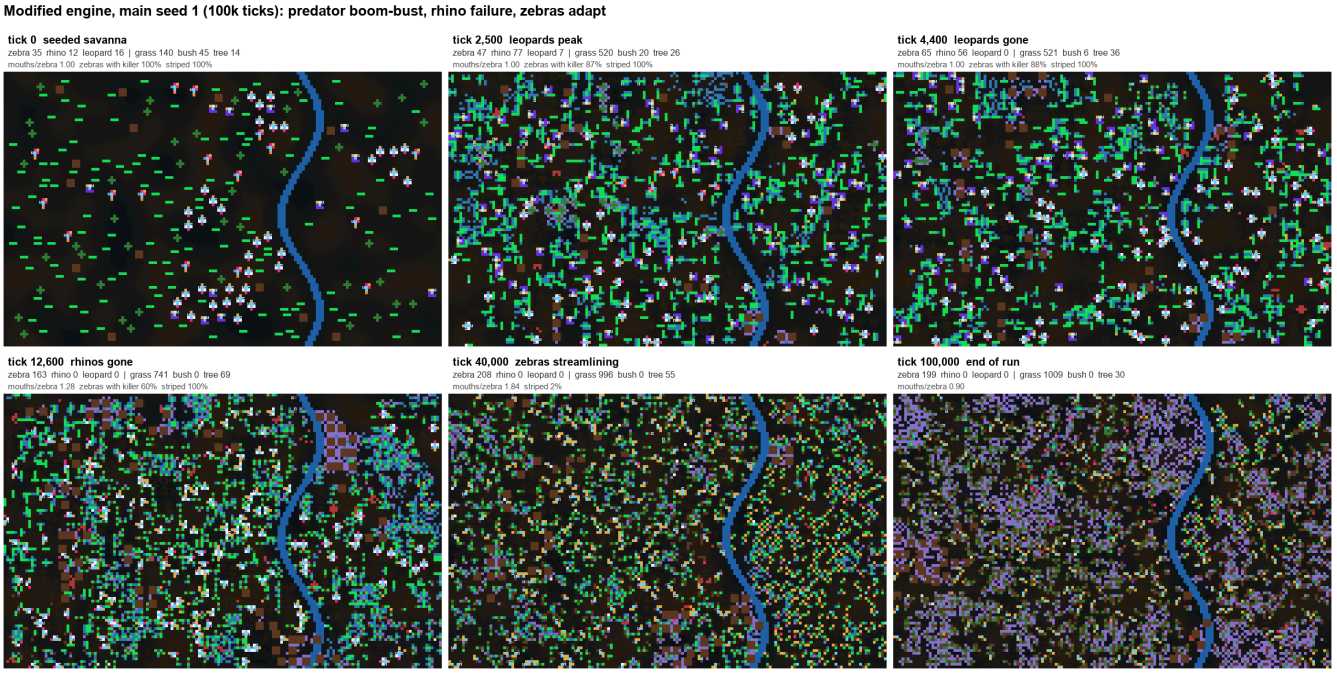
Target	Achieved?	What happened	Why
Zebra overgrazing suicide (our hypothesis)	No	Zebras evolved a second mouth (up to 2.8 per zebra) and survived where they did	Water-limited regrowth, the bush fallback and the river refuge kept grass food from collapsing. Where food did fall (200k-tick dry runs) it kept falling after zebras died, so plant crowding caused it. Capping zebras at one mouth made them die out more often, so the trait was adaptive
Rhino failure to adapt	Yes	Rhinos died out in every main seed, almost all of old age	At 16 food per calf they could not replace themselves once zebras took most of the grass
Zebra failure to adapt	Yes, where no second mouth evolved	Herds faded over tens of thousands of ticks with food still on the map	One-mouthed zebras barely paid their hunger cost and rarely bred

Target	Achieved?	What happened	Why
Leopard collapse	Yes, but not suicide	Leopards boomed and crashed in every seed	With 16 they aged out or were kicked to death before prey crashed. With 48 they ate every zebra and starved: over-exploitation set up by the starting density, not an evolved trait
Horn-trampling suicide	No	Rhinos never became common enough	Rhinos stayed rare, so trampling stayed small
Fire-driven collapse	No	Even 10 times the fire chance burnt at most 160 cells per fire	Patchy plants on mostly wet soil gave fires no connected fuel

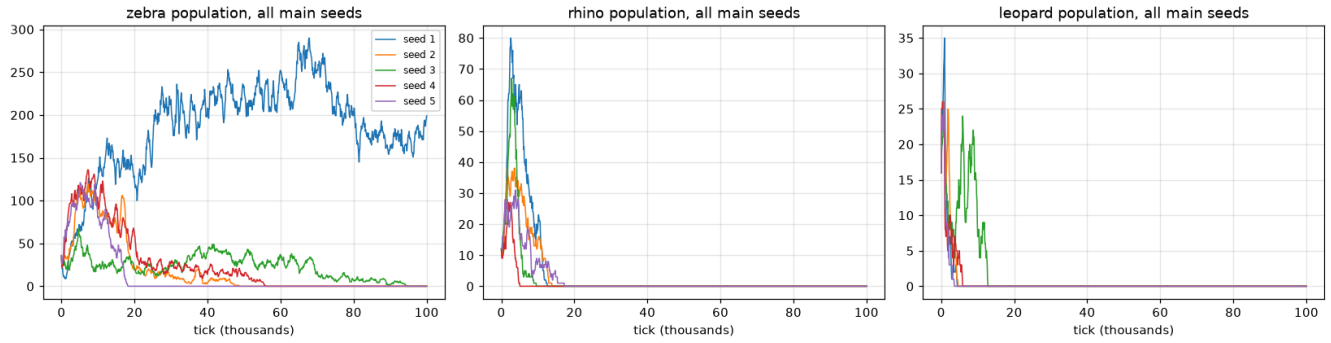
Compared with Part 1: the modifications closed the road the base engine used (killers destroying plants and herds) and opened the grazing road we wanted to test. That road needed grass to regrow slower than it was eaten, and in every setting we tried, regrowth or the fallbacks kept up.

Modified engine: five 100,000-tick runs

In all five seeds, leopards boomed and died out within 13,000 ticks and rhinos faded out within 18,000. Zebras survived only where they evolved a second mouth.

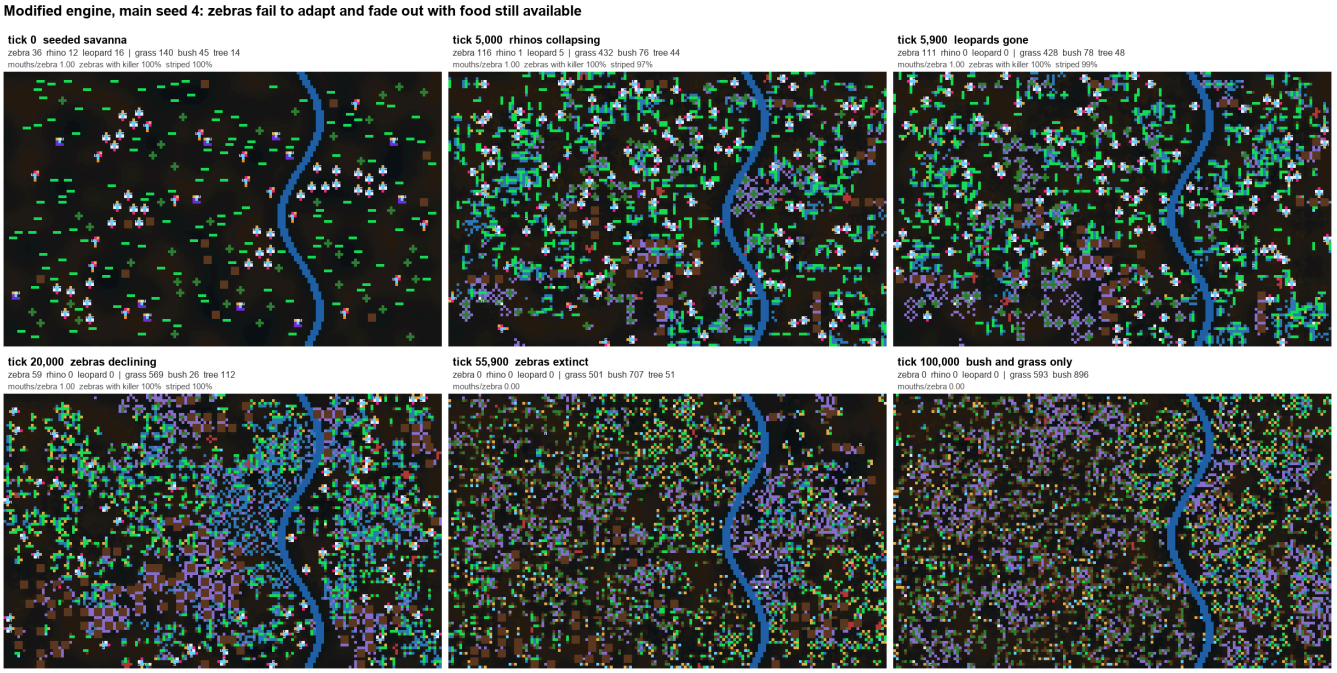


Seed	Leopards: peak, extinct at tick	Rhinos: peak, extinct at tick	Zebras: peak, outcome	Max mouths per zebra
1	35, 4,400	80, 12,600	290, survived (199 at tick 100k)	1.99
2	26, 4,600	38, 13,900	120, extinct at 48,700	1.10
3	26, 12,800	67, 9,600	67, extinct at 94,400	1.20
4	26, 5,900	27, 5,100	136, extinct at 55,900	1.00
5	26, 3,600	31, 17,300	121, extinct at 18,200	1.03



What happened to each species

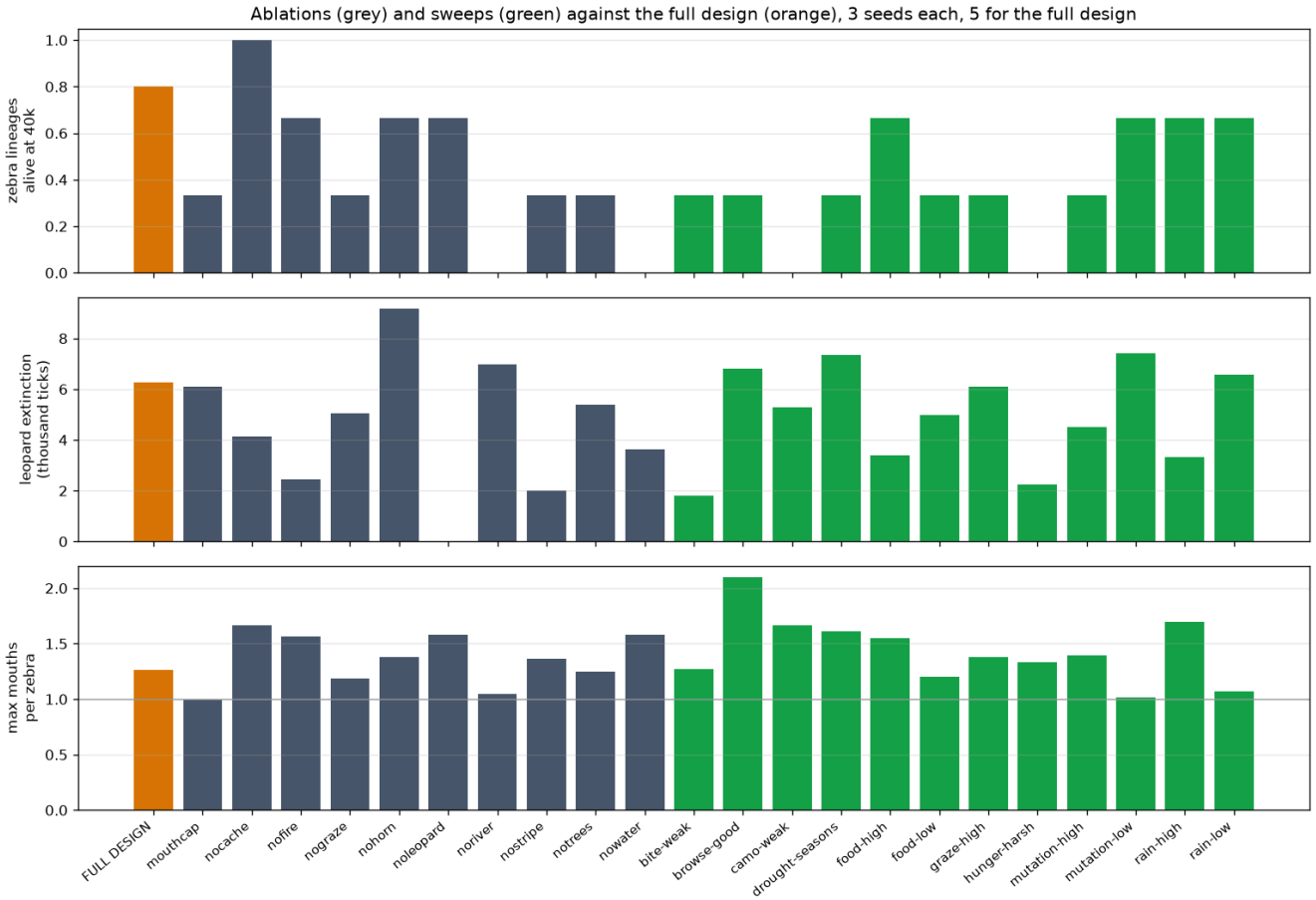
- **Leopards: boom then bust.** Every seed began with 16 leopards and they rose to 26 to 35 within about 1,000 ticks by eating the herd (57 to 144 zebras and 17 to 66 rhinos killed per run). They then died out. Old age was the main cause in seeds 1 and 3, as the first boom generation aged out without enough kills to replace itself. Zebra kicks were the main cause in seeds 2 and 4 (36 and 48 leopards killed). Caches were used 76 to 870 times per run, which delayed but never prevented extinction. Zebras did not crash before the leopards died, so this is a predator that failed to persist rather than over-predation.
- **Rhinos: failure to adapt.** Rhinos rose from 12 to 27 to 80, then died out. Nearly all deaths were old age (54 to 358 per run) with only 17 to 66 killed by leopards. At 16 food per calf they could not replace themselves once zebras took most of the grass. No horn-trampling collapse appeared.
- **Zebras: survival depended on adapting.** In seed 1, mouths per zebra rose from 1.0 to 1.99 and the herd stayed large. In seeds 2 to 5 the mouth trait did not spread. Zebras then declined slowly over tens of thousands of ticks, dying of old age (1,081 to 2,029 deaths) while food was still on the map. Starvation killed only 4 to 81 per run.
- **Plants and fire:** grass grew to between roughly 500 and 1,200 plants, and bushes expanded or vanished depending on how much grass took their water. Fires were frequent (29 to 41 per run) but small (154 to 381 cells burnt in total), so fire was a minor force.



Ablations and sweeps

The river, soil water and stripe camouflage were what kept zebras alive. Without the cache, leopards died sooner and every zebra lineage survived.

Each ablation switches off one feature, and each sweep changes one setting of the full design. Each ran for 3 seeds of 40,000 ticks. The reference is the five main runs read at tick 40,000, when 4 of 5 zebra lineages were alive. With 3 seeds, a one-seed difference is within noise. We treat only 0 of 3 or 3 of 3 as a clear signal.



Condition	Zebra lineages alive at 40k	Other notable result	Reading
Full design (reference)	4 of 5	Leopards extinct by about tick 6,300 on average	Baseline
No river	0 of 3 (median extinction tick 18,300)	Lowest zebra peak (84)	The riverbank refuge kept zebras going

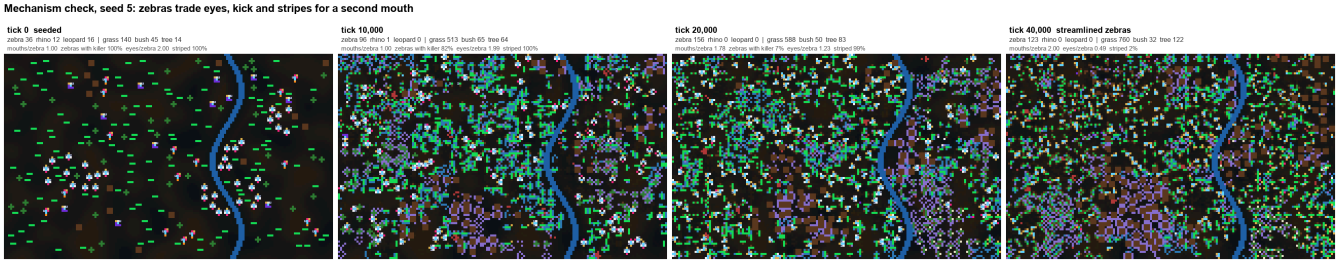
Condition	Zebra lineages alive at 40k	Other notable result	Reading
No soil water	0 of 3 (median tick 21,100)	Grass spreads unchecked	Water limits matter to zebras; the exact cause of their decline here was not isolated
No stripes	1 of 3 (median tick 11,300, the earliest)	Zebra peak 78	Herd camouflage protects zebras
Weak camouflage (maximum 30%)	0 of 3	Confirms the stripe result	Herd camouflage protects zebras
No cache	3 of 3	Leopards extinct sooner (about tick 4,100)	The cache prolongs predation, which costs zebras
No trees	1 of 3	Leopards (who could then cache anywhere) peaked at 110, and persisted in one seed	Tying the cache to trees limits leopards
No horn	2 of 3	Leopards lasted longest (about tick 9,200)	Horns shorten predator survival
Mouth cap (zebras limited to 1 mouth)	1 of 3	Mouths stayed at 1.00	Blocking the extra mouth harmed zebras: the trait is adaptive, not suicidal
No grazing	1 of 3	-	Grazing damage was not what killed zebras
No leopards	2 of 3	Zebra peak 201	Predators are not needed for zebras to fade
No fire	2 of 3	-	Fire is a minor force
High food production (8)	2 of 3	Highest zebra peak (225)	More food helps but does not guarantee survival
Low food production (3)	1 of 3	Rhinos survived in one seed	Scarcity favours slow, armoured rhinos

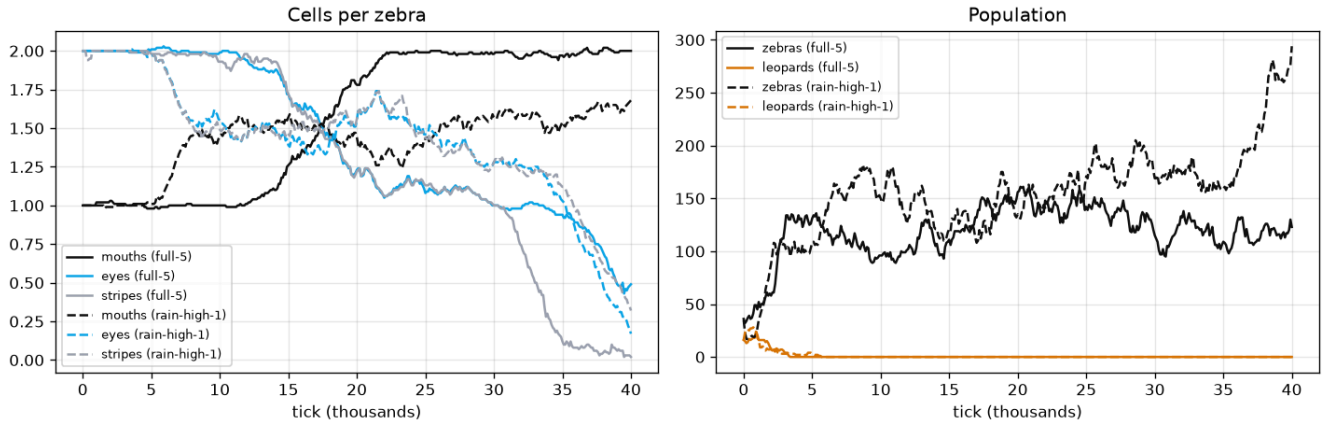
Condition	Zebra lineages alive at 40k	Other notable result	Reading
High or low rain	2 of 3 each	High rain gave mouths 1.70	Rain alone did not decide the outcome
Seasonal drought	1 of 3	-	Within noise
High grazing damage	1 of 3	-	Within noise
Good bush food (0.7)	1 of 3	Highest mouths (2.10)	A better fallback did not rescue zebras
Harsh hunger (6 missed payments)	0 of 3	Leopards extinct by about tick 2,200	Energy stress kills grazers and predators alike
Weak bite (damage 2)	1 of 3	Leopards extinct by about tick 1,800	Leopards need a strong bite to survive at all
High or low mutation	1 of 3 and 2 of 3	Low mutation kept mouths at 1.01	Faster change did not help zebras

Mechanism check: zebras streamline

When leopards disappear, zebras that survive swap costly defensive cells for a second mouth. This is an adaptation, not a suicide.

The main logs recorded mouths but not other cell types, so we added per-cell counts and reran 8 runs (5 seeds with high rain, 3 with default settings) with screenshots every 5,000 ticks.

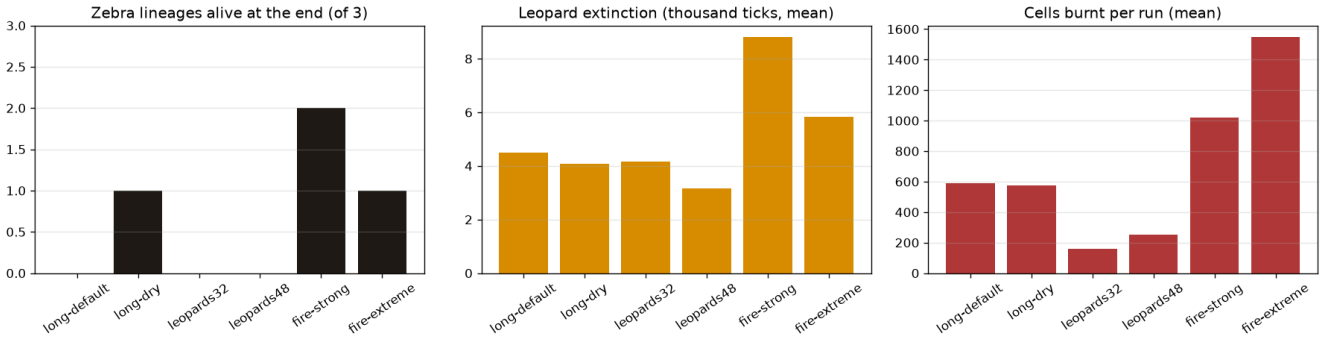




- In default seed 5, eyes per zebra fell from 2.0 to 0.49, stripes from 2.0 to 0.02, and the kick from 1.0 to 0 by tick 25,000. Mouths rose to 2.0, and the herd held at about 110 to 150.
- In high-rain seed 1, the same shift took the body from 8 to 4.3 cells, and the herd grew to 294 by tick 40,000.
- Once leopards are gone, eyes, stripes and the kick are dead weight. A smaller body needs less food per calf, and a second mouth gathers food faster, so the streamlined lineage wins.
- In the four high-rain seeds where zebras died, the shift never started. Mouths stayed at 1.0 and zebras faded out by tick 10,000 to 30,000 with food still present. That is failure to adapt.
- Grass food did not crash under two-mouthed zebras: it stayed between roughly 500 and 2,000 cells. Water-limited regrowth plus the bush fallback kept the extra-mouth trait from removing its own food, which is the third criterion our hypothesis needed.

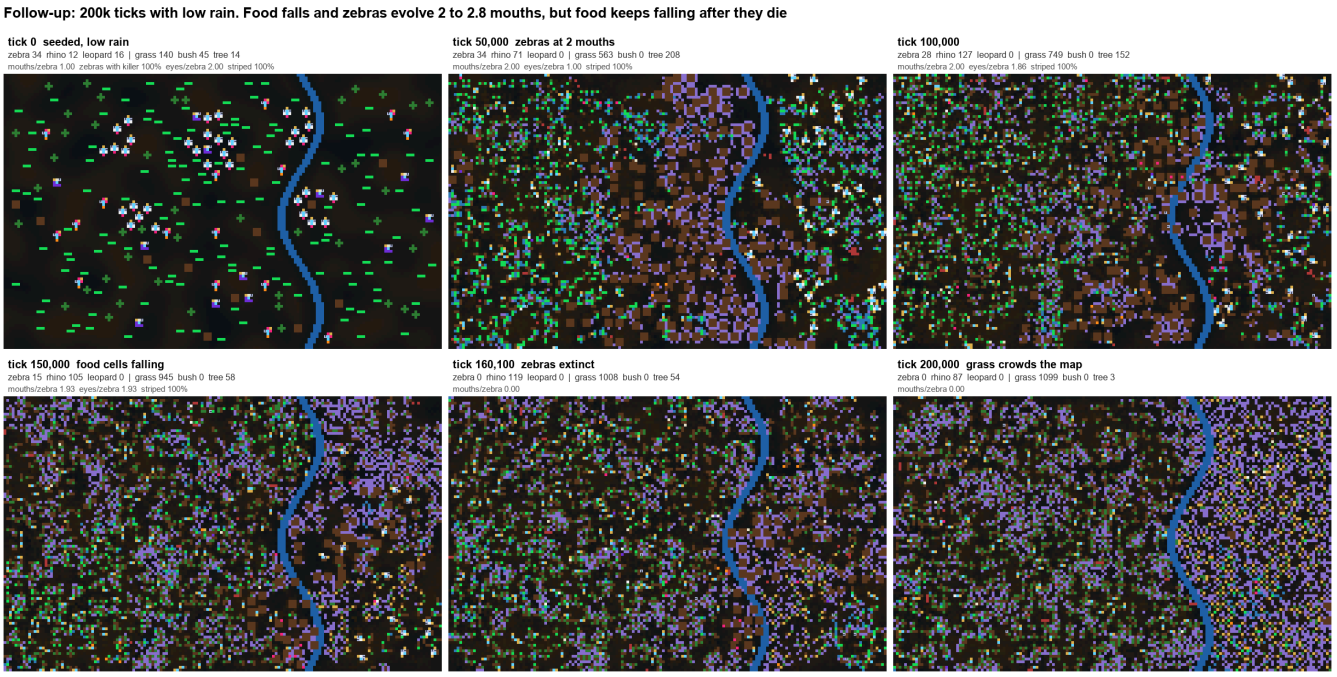
Follow-up runs: longer, drier, more predators, stronger fire

Eighteen more runs (3 seeds each) tested the three open questions. None produced zebra overgrazing suicide or a persistent predator. Stronger fire burnt more but stayed patchy.



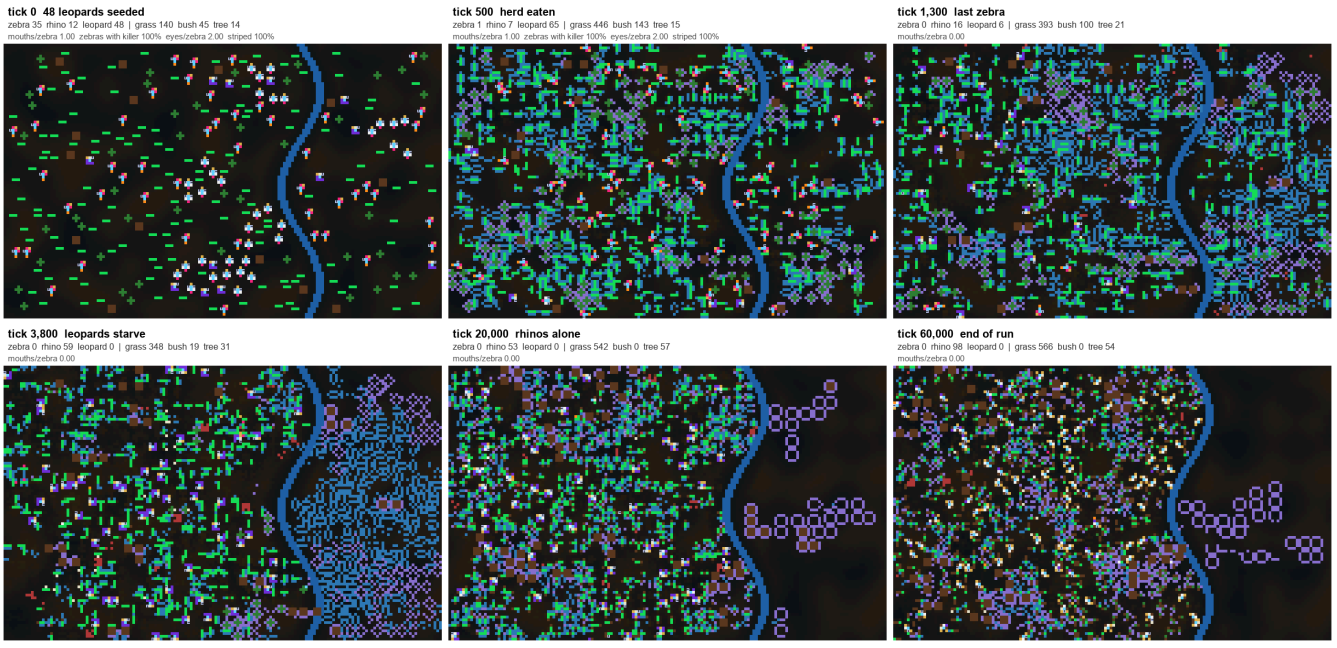
Condition	Settings	Zebras	Leopards	Fire
Long default	200k ticks	0 of 3 survived; mouths rose to 2.0 in 2 seeds	Extinct by tick 2,400 to 7,400	About 590 cells burnt per run
Long dry	200k ticks, rain 0.0008 (40% of normal)	1 of 3 survived; mouths up to 2.8	Extinct by tick 2,000 to 7,100; in one seed the zebras also crashed (over-predation)	About 570 cells per run
32 leopards	Twice the starting predators, 60k ticks	0 of 3 survived (extinct by tick 17,700 to 47,000)	Peaked at 44 to 48, extinct by tick 3,200 to 5,800	Small
48 leopards	Three times the starting predators, 60k ticks	Herd eaten within about 1,300 ticks in 2 seeds	Peaked at 66 to 70, starved by tick 2,000 to 3,800	Small
Strong fire	Fire chance 5x, spread 1.0, up to 4,000 cells	2 of 3 survived at 40k	Extinct by tick 3,500 to 15,600	About 1,020 cells per run, largest single fire 160, 270 to 340 plants killed
Extreme fire	Fire chance 10x, up to 8,000 cells	1 of 3 survived at 40k; mouths 2.0 in all seeds	Extinct by tick 3,300 to 10,900	About 1,550 cells per run, largest single fire 107, 370 to 520 plants killed

Longer and drier runs: zebras adapt to scarcity they did not cause



In long dry seed 3, grass food fell from about 1,550 cells at tick 20,000 to 166 at tick 160,000, and zebras answered by evolving 2 to 2.8 mouths. They still died out at tick 160,100. Food kept falling after the last zebra died (to 36 cells at tick 190,000), and grass plants doubled over the same period. So the food shortage came from grass crowding the map, not from zebra grazing. The extra mouth was a response to scarcity, not its cause, so this does not meet criterion 3. Doubling the run length did not change the conclusion.

More leopards: over-exploitation, not persistence



With 48 leopards the herd of 35 was gone by tick 1,300, and the leopards starved by tick 2,000 to 3,800. This is the textbook predator over-exploitation collapse: the predators destroyed their only prey and then died. It is ecological, not evolutionary. It came from the starting density we chose, with no trait spreading first, so it is not evolutionary suicide. With 32 leopards the boom was larger (peaks of 44 to 48) but ended just as fast. More predators only shortened the system's life.

Stronger fire: still limited by patchy fuel

Raising the fire chance 10 times raised burning only about 2.6 times (from about 590 to 1,550 cells per run), and no single fire passed 160 cells. Plants in this world grow as scattered small organisms on mostly wet soil, so fires run out of connected dry fuel. Fire killed hundreds of plants but did not change who survived beyond seed-to-seed noise.

Answers to the brief

1. Did an extinction event occur? Yes. Zebras died out in every base condition, and every leopard and rhino population died out in the modified main runs, as did zebras in 4 of 5 main seeds.

2. Mis-adaptation or failure to adapt?

- **Mis-adaptation (evolutionary suicide):** base-engine plant-kill control, seed 2. A killer evolved, spread to 89-100% of zebras because every plant or zebra it killed became food, destroyed all the plants, and the zebras died out. Seeds 1 and 3 collapsed on their own too but kept their grass (3 of 4 criteria).
- **Self-destruction by design, not evolution:** every base condition where zebras started with a kick. The kick destroyed the plants and the herd, but it was seeded, so it fails the trait-spread criterion.
- **Failure to adapt:** base zebras with mutation off, base leopards (no speed advantage), modified rhinos in every seed, and modified zebras where the second mouth never evolved.
- **Not observed:** our main hypothesis, zebras overgrazing with extra mouths and falling back on poor bush food. The extra mouth spread (up to 2.8 per zebra) but never removed its own food.

3. Environment versus seeded species. The environment decided which road was open. In the base engine, killers could destroy plants and herds, so that is how populations collapsed. In the modified engine that road was closed and grazing became possible, but water regrowth, fallback food and the river kept zebras from eating themselves out. Within the modified savanna, removing the river or soil water was as decisive as removing a trait (0 of 3 zebra lineages alive at 40k for each).

4. Instrumental features and traits.

- **Toward extinction:** killer cells that hurt plants and kin (base engine), costly armour (slow rhino breeding), leopards' single-prey diet, harsh hunger.
- **Away from extinction:** the river refuge, soil-water regrowth, stripe camouflage, the zebra kick against leopards, and the evolved second mouth with a streamlined body.
- **Cache:** delayed leopard collapse but never prevented it; its main effect was to keep predation on zebras going longer.

Limitations

In the base engine, the evolved killer collapsed the zebras in 3 of 3 seeds, but only 1 seed met all four criteria. The modified-engine results rest on settings we tuned by trial runs and on 3 seeds per condition.

- **Tuning history.** Before the final runs, we changed rules after seeing results, because early designs failed for mechanical reasons:
  - Leopards were redesigned three times: the killer was on the side, the eye blocked the jaws, and they could not eat their kills.
  - Sprinting was limited to chasing and fleeing, because fast zebras trampled their own food.
  - Armour cost and health were raised.
  - Weapons were limited to striking predators, after zebra kicks were found killing rhinos.

  All of these are general rules about body parts, not species switches. They were fixed before the final matrix, and every final run used the same rules.
- **Small samples.** Ablations and sweeps have 3 seeds, so a one-seed difference is noise.
- **Different run lengths.** Main runs ran for 100k ticks, ablations and sweeps for 40k. We compare them at tick 40k.
- **Not exactly repeatable.** Seeds fix the starting world but not the engine's random numbers, so reruns give different trajectories with the same overall pattern.
- **Arbitrary thresholds.** The classifier uses 1.5 times growth, a 0.4 mouth rise, a 0.3 killer rise and a 25% plant fall. The per-run evidence is in `output/summary.json` so other thresholds can be checked.
- **Lineages are labelled by their seeded ancestor.** The "grass" lineage evolved mouths (about 1 per plant by tick 100k) and a few mobile forms, but it is still counted as grass.
- **Weak fire.** Fires burnt at most a few hundred cells per run, and even 10 times the fire chance (follow-up runs) gave fires of at most 160 cells, so fire stayed a minor force.

- **Small leopard population.** With 16 leopards on a 16,000-cell map, they failed to persist in every main seed. Doubling or tripling them (follow-up runs) made them crash faster, so long-term predator-prey cycles were never observed.
- **Base engine is not completely stock.** The base runs include our Assignment 2 change that lets plants reproduce from producing food. Without it, plants without mouths never reproduce in the stock engine.
- **Base runs were redone with one-mover species.** Our first base runs gave zebras and leopards two movers each. After checking the engine creator's explanation that extra movers add nothing, we reran all 18 base runs with one mover each. The results in Part 1 come from the reruns; the earlier runs are discarded.
- **The base zebra's kick dominates Part 1.** We gave the base zebra a rear killer as a kick, as in our group's original design. In the original engine that kick hurts plants and other zebras, so it drove most base collapses. A base zebra without the kick would mainly test the plant-kill control route.

Reproducing the results

Everything is in the repository under `savanna-experiments/`, and the engine changes are under `src/`.

Step	Command or file
Build the engine	`npx webpack --mode=development`
Generate a world	`node savanna-experiments/make-world.js <variant> <seed>` (variants: `full`, `base-*`, `no*`, `mouthcap`)
Serve the engine and save outputs	`python savanna-experiments/server.py 8138`, then open `http://localhost:8138`
Run the experiment matrix	In the browser console, load `/exp/harness.js` and `/exp/jobs.js`, then call `Savanna.queue(JOBS)`
Charts and summary table	`python savanna-experiments/analyze.py` writes `output/charts/` and `output/summary.csv`
Report figures	`python savanna-experiments/make_figures.py` writes `output/figures/`

Step	Command or file
Observation log (state at every screenshot)	`python savanna-experiments/obs_log.py` writes `output/observation-log.md`

Every run folder in `output/` holds `log.csv` (every 100 ticks), `events.json` (collapses, extinctions and fires) and its screenshots. The worlds can also be loaded in the normal Life Engine interface through Load, then World.